# MEGN 441
# Intro to Robotics


Syllabus, Fall 2026

Hello! I am **Prof. Gary Nave**, and welcome to MEGN 441 Introduction to Robotics. I am a Teaching Faculty member in the Mechanical Engineering Department, and I am very much looking forward to learning together throughout this course. This is a class that has been recently re-designed, so your feedback is always welcome to help improve it for future students.

## Instructor Contact Information

Prof. Gary Nave (he/him)
Assistant Teaching Professor
Mechanical Engineering

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(Inside 280-K, in the East part of Brown)
Office Hours: See Canvas

## Class Details

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| Lectures : | MW 9:00-9:50, | Alderson 130 |
| | 10:00-10:50 | Alderson 130 |
| Labs: | M 11:00-1:50, | Brown 136 |
| | 2:00-4:50 | Brown 136 |
| | W 1:00-3:50 | Brown 136 |

## Course Description

**In MEGN 441, we build robots for good**. This hands-on, project-based class provides a gateway to the world of intelligent machines. We'll explore key ideas in robotics, including the dynamics of robot arms, perception of obstacles, and planning and control of motion for mobile robots. We'll program mobile robots through our semester-long lab sequence using the Robot Operating System (ROS2), which is used throughout both academic and industrial robot development. Specifically, students will use key ROS2 packages such as: rviz, gazebo, slam_toolbox, nav2, and more. We will learn all of this in an active, welcoming learning environment that emphasizes both technical and professional skills.

This course is open to all majors at Mines and is designed to be welcoming and accessible to those new to robotics while also being challenging enough for those who have been working with robots for years. Through our lecture components, we will derive the actual motion of a robot based on the motion of its actuators, improve the performance of their robots using motion planning algorithms, and analyze the ethical implications of robotics. Through our labs, we will work together in teams to create robots that incorporate perception, kinematics, and navigation to deliver packages to target locations.

### Statement on Classroom Community & Support

One of my core beliefs related to education is that every student that comes into my class has the capacity to succeed, no matter how they've been told in the past that they don't belong. It is essential to me that everyone feels welcomed and included in this class. I have designed the class to be both supportive of students new to robotics and challenging for those who have been working with robots for years. Because students enter this class with a wide range of background knowledge in robotics, mechanical systems, electronics, and programming, this is a class where there will be plenty of opportunities to learn from and with one another. Some strategies that we will use to make this course more inclusive, accessible, and supportive of students are:

- Regular small-group discussions and peer-to-peer interactions within the lecture time to build community within the classroom, as well as peer-to-peer interactions within lab teams.
- Regular opportunities to share feedback with your professor about how the course is going.
- Peer learning opportunities, where student teams will provide a brief (5-minute) presentation on a skill or topic related to robotics that they know or have researched in our lab section.
- Structure to provide support for students to make progress with the freedom for students to explore whatever approach makes sense to them.
- Course assignments intended to get students to engage with societal and ethical issues around robotics, rather than just focusing on the technical aspects as if they are unrelated.

It is incredibly important to me that you feel like a valued member of this classroom community, so please reach out to me if you come across anything related to this class that is implicitly or explicitly exclusive or inaccessible. Thanks for being here!

## Course Learning Outcomes (CLOs)

1. **Programming** - Program a physical mobile robot using the Robot Operating System (ROS2) using Python or C++ and common ROS2 packages such as nav2, gazebo, and more. *Assessment: Exam 1 + All labs, particularly lab 1*
2. **Perception** - Integrate sensors, such as depth cameras and LiDAR, to identify obstacles and view using Rviz. *Assessment: Labs 2 and 4, particularly lab 2*
3. **Kinematics** - Derive the forward and inverse kinematics of open-chain robotic arms using homogeneous transformation matrices. *Assessment: Exam 2 + Labs 3 and 4, particularly 3*
4. **Navigation –** Apply navigation algorithms for a robotic system to choose a path to reach a target. *Assessment: Exam 2 + Lab 4*
5. **Debugging –** Improve the performance of robotic systems by managing system parameters and recording data for analysis. *Assessment: Exam 1, and all labs*
6. **Ethics** - Analyze the ethical implications of the use of robotic systems to solve problems. *Assessment: Reading assignments and robot presentations*

7. **Overall** - Create a robot that uses perception, planning, and navigation to accomplish a goal, both in simulation and in practice. *Assessment: All labs, especially labs 4-5*

## Summative Assessments

**Exams (24% total)**

Two exams during the semester will be used to assess understanding of key topics. Exams will be open-note, but not open internet. Any course materials or pre-prepared notes are welcome to be used.

1. **ROS2 fundamentals –** Students will be tested on whether they understand key ROS topics. These are used through the labs, but it is also important to test each student's understanding. Topics will include: ROS2 Nodes, topics, actions, services, and parameters, Linux command line tools, debugging using ros bags
2. **Robot kinematics and navigation –** Students will be tested on forward and inverse kinematics of open-chain robotic arms and homogeneous transformation matrices, as well as fundamental navigation algorithms. These will be used in controlling the robot arm and navigating the robot, but it is important for students to demonstrate their comfort with the mathematics of robot kinematics and navigation algorithms directly.

**Lab Sequence (50% total)**

For each lab in the sequence, students will submit their code repository, a video of the robot accomplishing its goal both in simulation and in practice, and a writeup of the robot's control design including a reflection on their learning in each lab. Each lab will be accessed on the quality and efficiency of the code, the success in accomplishing each task, and the clarity and thoughtfulness of the writeup.

1. **Introduction to Rosbots, ROS2 Packages, and Remote Control**: Students will be tasked with programming a robotic car to drive a prescribed course using ROS2, using Gazebo and a physical system, navigating a series of waypoints. Students will measure the accuracy of different internal tracking systems.
2. **Simultaneous Localization and Mapping (SLAM)**: Students will be tasked with using a depth camera and a 2D Lidar to identify obstacles in a room and generate a map of the room as the car drives around, using a Simultaneous Localization and Mapping (SLAM) algorithm.
3. **Robot Arm Pick and Place**: Students will apply inverse kinematics to control the gripper on a 5R robot arm, located on top of the robotic car, using computer vision to identify and lift an object.
4. **Delivery and Navigation**: Students will be tasked with combining the lessons from the first three labs to design a car that can map a space and navigate around obstacles to deliver an object to a target location.
5. **Final Project Showcase**: As a final showcase of the skills developed during the semester, students are tasked with diving deeper into a topic in robotics, either expanding their existing Rosbot capabilities or develop and building a new mechatronic or robotic system.

## Formative Assessments

**Ungraded formative assessments**

1. For each lab, students will complete a reflection on their group dynamics, roles, and their own efforts within the group. Groups will be switched for each lab, so students will grow in their ability to work with other students.
2. At the beginning of the semester, students will complete a background knowledge check-in, coupled with a similar check-in near the end of the class. Results will be shown to demonstrate to students how far they have come. Students come from different majors, so the background knowledge check-in will be very helpful.
3. List of topics of interest – students will be asked in-class to complete a brief survey of robotic topics of interest. Part of the goals of an introductory course is to show all of the different areas of robotics. Students will reflect on which parts of robotics they find most interesting.

**Homework (10% total)**

Homework assignments will provide the opportunity to practice and apply key concepts from lecture. Some will be in the form of a programming assignment, with code submission and a write-up, while others will be more focused on basic principles of robotics, such as control, kinematics, or navigation.

Homework assignments will be accepted late, with a penalty of 25% per day.

**Reading assignments (6% total)**

Every few weeks, students will choose from short list of key articles to read and engage in discussion both on Hypothesis and in-class. In-class discussions will be written about (less than one page) and submitted as a group. For pre-class readings, late submissions will not be accepted. Reading assignments will be closed the day of class and then re-opened after they have been graded, so that students will continue to have access to them.

**Peer Learning Presentations (10% total)**

Students will have the option to present a researched presentation to their classmates on EITHER: a key robot that has been developed that presents an exciting development to the world of robotics OR a key topic around robotics chosen from a list (e.g., computer vision, reinforcement learning, swarm robotics, etc.). Students will give brief presentations in teams, providing a discussion of both the interesting technical methods used and any ethical considerations to the context of these robotic situations. Student presentations will be done to start the lab session each week and will involve both a presentation and some discussion from the class.

## Generative AI Policy

Generative AI use is **Selectively Permitted** in this class, meaning that I will communicate clearly when and how genAI tools are permitted. In general, I encourage you to reflect on whether you believe that generative AI will be helpful or harmful to your learning for a given assignment.

- On **lab assignments**, genAI use is permitted, although you are accountable for the work you do and are required to be transparent about genAI use. All lab assignments will have a genAI appendix for you to communicate your AI use.
- On **exams**, genAI use is strictly prohibited.
- On **homework assignments**, you may use genAI to support your own work, but you may not generate the entire assignment or upload the assignment or question directly. The homework assignments are designed for you to challenge yourself and apply your knowledge.
- **Peer learning presentations** may use genAI to assist research and collection of presentation materials, but all genAI use must be transparently communicated.

# Campus Resources + Policies

## Diversity and Inclusion:

At Colorado School of Mines, we understand that a diverse and inclusive learning environment inspires creativity and innovation, which are essential to the engineering process. We also know that in order to address current and emerging national and global challenges, it is important to learn with and from people who have different backgrounds, thoughts, and experiences.

Our students represent every state in the nation and more than 90 countries around the world, and we continue to make progress in the areas of diversity and inclusion by providing Diversity and Inclusion programs and services to support these efforts.

## Disability Support Services:

The Colorado School of Mines is committed to ensuring the full participation of all students in its programs, including students with disabilities. If you anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with me. Students with disabilities may also wish to contact Disability Support Services (DSS) to discuss options to removing barriers in this course, including how to register and request official accommodations. Please visit the DSS website for contact and additional information. If you have already been approved for accommodations through DSS, please meet with me at your earliest convenience so we can discuss your needs in this course.

Accessibility within Canvas: Read the Accessibility Statement from Canvas to see how the learning management system at the Colorado School of Mines is committed to providing a system that is usable by everyone. The Canvas platform was built using the most modern HTML and CSS technologies, and is committed to W3C's Web Accessibility Initiative and Section 508 guidelines.

## Discrimination, Harassment, and Sexual Misconduct:

Discrimination, Harassment, and Sexual Misconduct of any type, including sexual harassment, sexual assault, dating violence, domestic violence, and stalking, are prohibited under the Policy Prohibiting Unlawful Discrimination and the Policy Prohibiting Sexual Harassment, Sexual Assault, and Interpersonal Violence. As a participant in this course, we expect you to respect your instructor and your classmates. As your instructor, it is my responsibility to foster a learning environment that supports diversity of thoughts, perspectives and experiences, and honors your identities. I am also a mandatory reporter as an instructor at Mines, and if I receive a disclosure of Discrimination, Harassment, and/or Sexual Misconduct, I am required to report it to the Title IX Coordinator. You can also make a report if you are witness to or impacted by Discrimination, Harassment, and/or Sexual Misconduct If something is said or done in this course (by anyone, including myself) that made you or others feel uncomfortable, or if your performance in the course is being impacted by your experiences outside of the course, please report it to me (if you are comfortable doing so), to the ttps://www.mines.edu/institutional-equity-title-ix/reporting/, and to https://www.mines.edu/institutional-equity-title-ix/submit-report/ (an anonymous option). You can also contact the Mines Title IX Coordinator, Carole Goddard, directly at 303.273.3260 or titleix@mines.edu for more information. It's on us, all of the Mines community, to engineer a culture of respect.

## Preferred First Name Project:

In order to foster a more inclusive environment for students and faculty who use Canvas to teach, learn and collaborate, ITS is implementing the use of preferred first names throughout the entire Canvas learning management system at Mines on August 15, 2022. This change will transition Canvas to fully utilize preferred first names regardless of feature, function, or user. You can add a preferred first name by following the steps on the Preferred First Name website.

Why Are Preferred First Names important? Calling a person by their preferred name shows respect. Using preferred names contributes to the University's goal of providing an empowering, safe and nondiscriminatory educational and work environment. Someone's name is an extremely important part of a person's identity. We encourage everyone to utilize a person's preferred name and pronouns whenever addressing or referring to them.

## CARE @ Mines:

If you feel overwhelmed, anxious, depressed, distressed, mentally or physically unhealthy, or concerned about your wellbeing overall, there are resources both on- and off-campus available to you. If you need assistance, please ask for help form a trusted faculty or staff member, fellow student, or any of the resources below. As a community of care, we can help one another get through difficult times. If you need help, reach out. If you are concerned for another student, offer assistance and/or ask for help on their behalf. Students seeking resources for themselves or others should visit care.mines.edu.

Additional suggestions for referrals for support, depending on comfort level and needs include:

- CARE at Mines for various resources and options, or to submit an online "CARE report" about someone you're concerned about, or email care@mines.edu.
- Counseling Center – Visit the Counseling Center website or students may call 303-273-3377 to make an appointment. There are also online resources for students on the website. Located in the Wellness Center 2nd floor (1770 Elm St.)
- Health Center – Visit the Health Center website or students may call 303-273-3381 for an appointment. Located in the Wellness Center 1st floor.

- Colorado Crisis Services - For crisis support 24 hrs/7 days, either by phone, text, or in person, Colorado Crisis Services is a great confidential resource, available to anyone. http://coloradocrisisservices.org, 1-844-493-8255, or text "TALK" to 38255. Walk-in location addresses are posted on the website.
- Food and/or Housing - Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. Furthermore, please notify your professor if you are comfortable in doing so. This will enable your professor to provide resources that may be available.

All of these options are available for free for students. The Counseling Center, Health Center, and Colorado Crisis Services are confidential resources. The Counseling Center will also make referrals to off-campus counselors, if preferred.

In an emergency, you should call 911, and they will dispatch a Mines or Golden PD officer to assist.

## The Writing Center:

The writing center is a free academic support service available to all members of the campus community including undergraduate and graduate students. We can assist you at any stage of the writing process, from brainstorming to final revisions. You do not need a complete draft to make an appointment. Our consultants are experts in a variety of composition and communication fields, providing support as you work on projects such as lab reports, essays, collaborative papers, scholarly publications, thesis chapters, and oral presentations. Whether you are focusing on organization or sentence structure, the Writing Center can evaluate your individual needs and tailor each appointment so that you become a more effective and efficient communicator. The Writing Center is open Sunday-Friday for in-person and online appointments. To learn more about our services and to make an appointment, please visit the Writing Center website. For questions, please e-mail writing@mines.edu.

## Policy on Academic Integrity/Misconduct:

The Colorado School of Mines affirms the principle that all individuals associated with the Mines academic community have a responsibility for establishing, maintaining and fostering an understanding and appreciation for academic integrity. In broad terms, this implies protecting the environment of mutual trust within which scholarly exchange occurs, supporting the ability of the faculty to fairly and effectively evaluate every student's academic achievements, and giving credence to the university's educational mission, its scholarly objectives and the substance of the degrees it awards. The protection of academic integrity requires there to be clear and consistent standards, as well as confrontation and sanctions when individuals violate those standards. The Colorado School of Mines desires an environment free of any and all forms of academic misconduct and expects students to act with integrity at all times.

Academic misconduct is the intentional act of fraud, in which an individual seeks to claim credit for the work and efforts of another without authorization or uses unauthorized materials or fabricated information in any academic exercise. Student Academic Misconduct arises when a student violates the principle of academic integrity. Such behavior erodes mutual trust, distorts the fair evaluation of academic achievements, violates the ethical code of behavior upon which education and scholarship rest, and undermines the credibility of the university. Because of the serious institutional and individual ramifications, student misconduct arising from violations of academic integrity is not tolerated at Mines. If a student is found to have engaged in such misconduct sanctions such as change of a grade, loss of institutional privileges, or academic suspension or dismissal may be imposed. The complete policy can be found in the Mines' Policy Library.